

MUHAMMAD NAJEEB BROHI

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Highly skilled and results-driven AI & Python Software Engineer with deep expertise in Large Language Models (LLMs), Machine Learning pipelines, Generative AI, and Advanced Data Scraping. Proven track record of architecting scalable Hybrid RAG (Retrieval-Augmented Generation) systems, developing robust API integrations, and automating complex enterprise workflows. Exceptionally capable of solving intricate backend problems, engineering predictive Machine Learning algorithms, fine-tuning custom AI models, and transforming raw data into high-level intelligent applications. Adept at working independently to engineer bug-free, highly optimized software architectures that drive business innovation.

TECHNICAL EXPERTISE

AI & Machine

Learning:

Predictive Machine Learning, Supervised/Unsupervised Learning, LLM Development, Scikit-Learn, TensorFlow, PyTorch

RAG & AI

Frameworks:

Hybrid RAG Pipelines, Vector Databases (Pinecone/Chroma), LangChain, Prompt Engineering, Semantic Search

Programming &

Backend:

Python (Advanced core & OOP), RESTful API Development, Integration & Architecture, Data Structures

Automation &

Scraping:

Advanced Web Scraping, Selenium, Workflow Automation, Dynamic Data Extraction, Anti-bot bypass

Data & Deployment: Pandas, NumPy, Matplotlib, Streamlit, Git/GitHub, Linux CLI

PROFESSIONAL EXPERIENCE

AI & Python Software Engineer Intern | *Gexton Institute of Technology*

- Engineered and deployed custom AI-driven automation bots, successfully reducing manual data entry and repetitive operational workflows by over 80%.
- Architected end-to-end backend pipelines integrating predictive Machine Learning models, third-party APIs, and modern Python libraries to optimize complex business logic.
- Spearheaded the design of intelligent Python-based web scrapers, extracting thousands of unstructured data points daily to feed into centralized analytical databases.
- Collaborated with senior teams to implement Generative AI capabilities, refining prompt architectures and seamlessly integrating intelligence into existing software ecosystems.

KEY PROJECTS & ARCHITECTURES

Enterprise Hybrid RAG System | *Python, LLMs, Vector Databases, LangChain, APIs*

- Architected a highly scalable Hybrid Retrieval-Augmented Generation (RAG) pipeline capable of ingesting, chunking, and querying massive proprietary document datasets.
- Integrated dense and sparse retrieval mechanisms with state-of-the-art LLMs to deliver context-aware, highly accurate, and hallucination-free automated responses.
- Designed custom API endpoints for seamless communication between the front-end user interface and the underlying AI-powered vector search engine.

Intelligent Financial Advisory Engine | *Machine Learning, Pandas, Predictive Modeling*

- Developed a complex, data-driven financial analysis engine that utilizes advanced machine learning algorithms to evaluate user income structures, budgets, and market indicators.
- Implemented predictive logic optimization to automatically generate highly customized wealth-management and investment strategies based on dynamic risk profiles.

AI-Powered Attendance & Monitoring Architecture

| *Computer Vision, Python, Streamlit, API Integration*

- Engineered a robust, real-time attendance tracking platform utilizing AI integration and dynamic QR-based data flow.
- Built an interactive, full-stack monitoring dashboard with dedicated secure portals for distinct user roles, executing real-time analytics and automated report generation.

Advanced Data Extraction & Automation Toolkit | *Selenium, Scrapy, High-Concurrency Python*

- Created a suite of high-performance automation micro-tools and scraping bots engineered to navigate complex dynamic sites, bypass restrictions, and extract targeted data.
- Structured raw extracted data streams using Pandas and NumPy, feeding sanitized datasets directly into machine learning training pipelines and BI dashboards.

EDUCATION

Intermediate Govt. Boys Higher Secondary School Garelo, Larkana	Completed 2026
Matriculation Govt. Boys Higher Secondary School Garelo, Larkana	Completed 2025